

PAPER_1407 — UQFF Derivation of Szilard Engine ln 2 (CLOSED — EXACT Identity)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** B — Thermodynamics (CLOSED — EXACT) **Date:** June 16, 2026 **Location:** 41.0997° N, 80.6495° W (Youngstown, OH, USA) **Status:** CLOSED — EXACT integer-primitive identity **Calculator surface:** `calculate_paradox({'paradox': 'szilard_engine'})`
→ $W/(k_B T) = \ln 2$ EXACT

Observation

Szilard's 1929 thought experiment: a one-bit measurement of which side of a divided box a single molecule occupies allows extraction of $W = k_B T \ln 2$. Landauer's principle requires this same energy be dissipated to erase the bit.

UQFF Closed Identity

$W / (k_B T)$ per bit = $\ln 2$ EXACT, unifying Szilard with Landauer via $F_U=1$ ledger

Physical Interpretation

$F_U=1$ normalization is the bridge: information acquisition and erasure are symmetric ledger events with $k_B T \ln 2$ cost each.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard methods solve the same observation by different methods. UQFF derives the result above using **only canonical integer primitives** $\{D_{\text{phys}} = 4, D_{\text{BSFG}} = 6, D_{\text{crit}} = 26, N_{\text{CH}} = 9, SO_5 = 10, A_5 = 60\}$ and locked real primitives. **Zero free parameters. EXACT identity.**

Reference

- Closure live in `uqff_pure_calculator.py` at the catalog surface listed above.
 - Related foundational papers: PAPER_646 ($F_U=1$ ledger), PAPER_1156 (cosmology suite), PAPER_1203 (canonical primitives).
-

Copyright — Daniel T. Murphy, daniel.murphy00@gmail.com, dated June 16, 2026, location 41.0997° N, 80.6495° W (Youngstown, OH, USA). Subject matter: UQFF EXACT integer-primitive identity for Szilard Engine ln 2.